

Spatio-Temporal Analysis and Modeling of Bike-Sharing Systems in London and Taipei

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Abstract—This work presents a comprehensive spatio-temporal analysis of large-scale bike-sharing data from London and Taipei. The study investigates both temporal dynamics and spatial structure of bike usage by analyzing hourly demand patterns, seasonal trends, and station-level variations. Temporal modeling is performed using feature-engineered regression and tree-based methods to capture daily and weekly demand behavior. Spatial analysis is conducted through clustering techniques and spatial statistics, including Moran’s I, Geary’s C, and Local Indicators of Spatial Association (LISA), to identify demand hotspots and spatial dependencies across stations.

Results show that bike demand exhibits strong daily seasonality with distinct peak-hour patterns in London, while Taipei demonstrates smoother and more uniformly distributed usage. Spatial analysis reveals significant local autocorrelation, indicating that station demand is influenced by neighboring activity rather than geographic location alone. Machine learning models further validate these findings, with linear models performing effectively for spatial prediction and nonlinear models capturing temporal variations.

Overall, the study highlights the interplay between temporal regularity and spatial dependence in urban mobility systems, providing insights into the structure and predictability of bike-sharing demand across different cities.

I. INTRODUCTION

Bike-sharing systems have become an important component of urban transportation networks, offering a flexible and sustainable mode of mobility for short-distance travel. The widespread deployment of such systems in major cities has led to the availability of large-scale usage data, providing an opportunity to understand travel demand patterns and user behavior at both temporal and spatial levels.

Analyzing bike-sharing data presents two key challenges. First, demand varies significantly over time due to factors such as commuting patterns, daily routines, and seasonal effects. Second, demand is not uniformly distributed across space, as certain locations consistently exhibit higher activity due to their proximity to commercial centers, transit hubs, or residential areas. These temporal and spatial dependencies make it necessary to adopt a combined analytical approach rather than studying each aspect in isolation.

In this work, we perform a comprehensive spatio-temporal analysis of bike-sharing data from two cities: London and Taipei. The analysis is structured into two main components. The temporal component focuses on identifying patterns in hourly demand, including daily seasonality and peak-hour

behavior, and applies machine learning models to predict demand using engineered time-based features. The spatial component examines station-level demand distribution and investigates spatial dependencies using statistical measures such as Moran’s I, Geary’s C, and Local Indicators of Spatial Association (LISA), along with clustering techniques to identify meaningful spatial groupings.

In addition to analyzing London data in detail, we include a comparative study with Taipei to highlight differences in demand behavior across cities. This comparison provides insight into how urban structure and usage patterns influence the nature of bike-sharing demand.

The contributions of this work are threefold. First, we demonstrate the presence of strong temporal regularities in bike usage, particularly driven by commuting patterns. Second, we show that bike demand exhibits significant spatial autocorrelation, with neighboring stations influencing each other more than geographic location alone. Third, we validate these observations through machine learning models, highlighting the differing roles of linear and nonlinear methods in capturing spatial and temporal dynamics.

Overall, this study provides a structured framework for understanding bike-sharing systems through combined temporal modeling and spatial data analysis, offering insights that can be useful for demand forecasting and urban planning.

II. DATASET DESCRIPTION

The analysis in this work is based on large-scale bike-sharing datasets from two cities: London and Taipei. Both datasets contain detailed trip-level records, including information about trip start times, end times, and station identifiers. These datasets enable the study of demand patterns across both temporal and spatial dimensions. The datasets were obtained from publicly available sources on Kaggle [1].

A. London Dataset

The London dataset consists of approximately 38 million trip records collected over multiple years. Each record includes attributes such as start and end timestamps, start and end station identifiers, and station names. In addition, a separate station metadata file provides geographic coordinates (latitude and longitude) for each station, enabling spatial analysis.

To facilitate analysis, the raw trip data is aggregated into two forms. First, system-level demand is computed by grouping trips at an hourly resolution, resulting in a time series representing total bike usage over time. Second, station-level demand is obtained by aggregating trips per station, allowing analysis of spatial distribution and station-specific activity. This aggregation significantly reduces data size while preserving meaningful demand patterns.

B. Taipei Dataset

The Taipei dataset, consisting of 8 million records, similarly contains large-scale trip records with timestamps and station information. However, unlike the London dataset, explicit geographic coordinates for stations are not available. As a result, the analysis for Taipei is limited to temporal patterns and station-level demand distribution, without detailed spatial analysis.

Temporal aggregation is performed by grouping trips at an hourly resolution to construct a time series of demand. Station-level demand is computed using station identifiers, enabling comparison of activity levels across stations.

C. Preprocessing

Preprocessing involves converting timestamp fields into a standard datetime format and extracting hourly time intervals for aggregation. Missing or invalid entries are removed to ensure data consistency. For temporal modeling, additional features such as lagged demand values and time-based attributes (e.g., hour of day and day of week) are derived from the aggregated time series.

For spatial analysis in London, station-level demand is merged with geographic coordinates to form a spatial dataset. Additional features, such as distance from the city center and neighborhood demand statistics, are computed to capture spatial structure and local dependencies.

These preprocessing steps transform the raw trip-level data into structured datasets suitable for temporal modeling, spatial analysis, and machine learning. The code [2] for the same is available in open source.

III. TEMPORAL ANALYSIS

A. Time Series Overview

To analyze overall demand behavior, the London dataset is aggregated at an hourly resolution to construct a system-level time series. This time series represents the total number of trips occurring in each hour across all stations.

The resulting time series reveals clear variations in demand over time, with noticeable fluctuations corresponding to daily activity cycles. Periods of low activity are observed during late night hours, while higher demand is concentrated during daytime and commuting periods. This aggregation provides a compact representation of large-scale trip data while preserving meaningful temporal patterns.

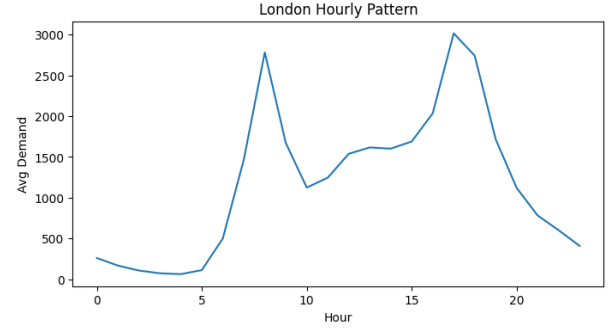


Fig. 1. Hourly bike-sharing demand in London, showing clear daily fluctuations and overall demand variation over time.

B. Seasonality and Trend Analysis

To further examine temporal structure, Seasonal-Trend decomposition using Loess (STL) is applied to the hourly demand series. The decomposition separates the time series into trend, seasonal, and residual components.

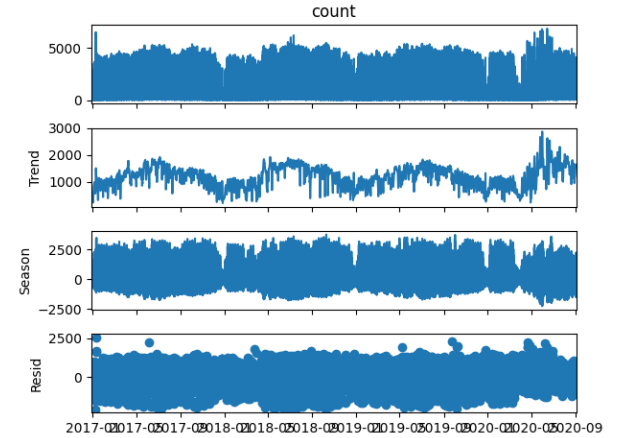


Fig. 2. STL decomposition of hourly demand showing trend, seasonal, and residual components. Strong daily seasonality is observed.

The seasonal component highlights a strong daily pattern, with consistent peaks during morning and evening hours. These peaks correspond to typical commuting times, indicating that bike usage is heavily influenced by work-related travel behavior. The trend component captures longer-term variations in demand, while the residual component represents irregular fluctuations not explained by the underlying structure.

These observations confirm that bike-sharing demand exhibits strong temporal regularity, making it suitable for predictive modeling using time-based features. Autocorrelation analysis further confirms the presence of strong periodicity, particularly at daily intervals.

C. Feature Engineering for Temporal Modeling

To enable predictive modeling, several features are derived from the aggregated time series. Lag-based features, such as demand in the previous hour, previous day (24-hour lag), and previous week (168-hour lag), are included to capture temporal

dependencies. Additionally, calendar-based features such as hour of the day, day of the week, and weekend indicators are incorporated to represent periodic patterns.

Rolling statistics, including moving averages and standard deviations over a 24-hour window, are also computed to capture short-term trends and variability. These features allow models to leverage both immediate past behavior and recurring temporal patterns.

D. Temporal Demand Prediction

Machine learning models are applied to predict hourly demand using the engineered features. A linear regression model is used as a baseline, while tree-based models such as Random Forest and XGBoost are employed to capture potential nonlinear relationships in the data.

Model performance is assessed using Root Mean Squared Error (RMSE) on a held-out test set.

TABLE I
RMSE COMPARISON FOR TEMPORAL MODELS WITH AND WITHOUT
FEATURE ENGINEERING

Model	Basic Features	Engineered Features
Linear Regression	456.859	4237.67
Random Forest	349.280	341.31
XGBoost	301.326	298.49

The results indicate that tree-based models outperform linear regression, demonstrating their ability to capture complex temporal interactions and nonlinear dependencies. Feature engineering improves performance for tree-based models, while linear regression is sensitive to feature scaling and multicollinearity.

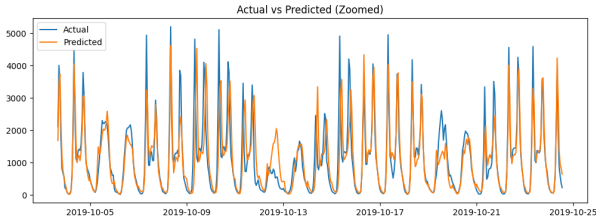


Fig. 3. Zoomed view of predicted vs actual demand, highlighting model performance at finer temporal resolution.

E. Model Evaluation

The predicted demand closely follows the actual demand trends, particularly during regular periods of activity. However, larger prediction errors are observed during peak hours, where demand spikes are more difficult to capture accurately.

Scatter plots of predicted versus actual values indicate that the models perform well overall, with predictions concentrated around the diagonal line. Minor deviations from the ideal line suggest underestimation of extreme demand values, which is typical in time series forecasting problems with sharp peaks.

Overall, the temporal analysis demonstrates that bike-sharing demand is highly structured and predictable, with

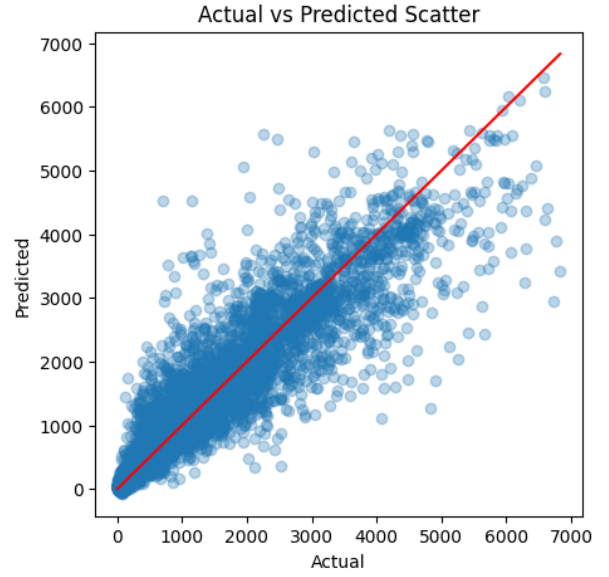


Fig. 4. Scatter plot of predicted versus actual demand values, showing strong overall agreement.

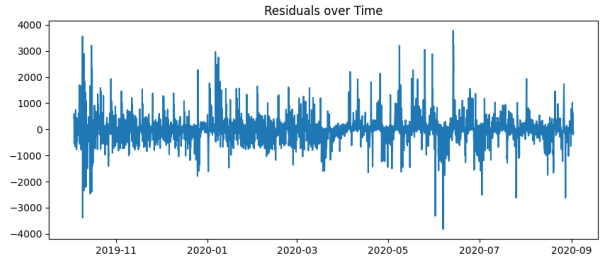


Fig. 5. Residual errors over time, indicating higher variance during peak demand periods.

strong daily seasonality and meaningful temporal dependencies that can be effectively modeled using feature-based machine learning approaches.

IV. SPATIAL ANALYSIS

A. Station-Level Demand Distribution

To analyze spatial patterns in bike usage, demand is aggregated at the station level by computing the average number of trips originating from each station. This aggregation provides a compact representation of spatial activity across the network.

The distribution of station-level demand reveals significant variation, with a small number of stations accounting for a disproportionately high volume of trips. These high-demand stations are typically located near central or high-traffic areas, while peripheral stations exhibit relatively lower activity. This uneven distribution suggests the presence of spatial structure in bike usage.

B. Spatial Autocorrelation

To formally evaluate spatial dependence, global spatial autocorrelation is measured using Moran's I and Geary's C statistics.

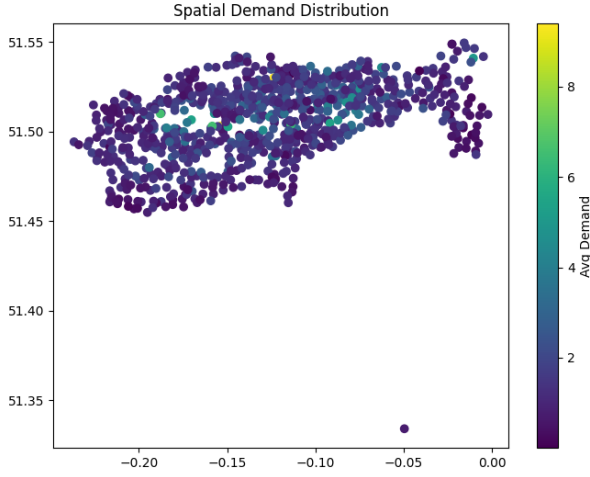


Fig. 6. Spatial distribution of station-level demand across London. High-demand stations are concentrated in central regions.

Moran's $I = 0.237$ ($p < 0.01$), indicating significant positive spatial autocorrelation. Geary's $C = 0.765$ ($p < 0.01$), further confirming non-random spatial structure.

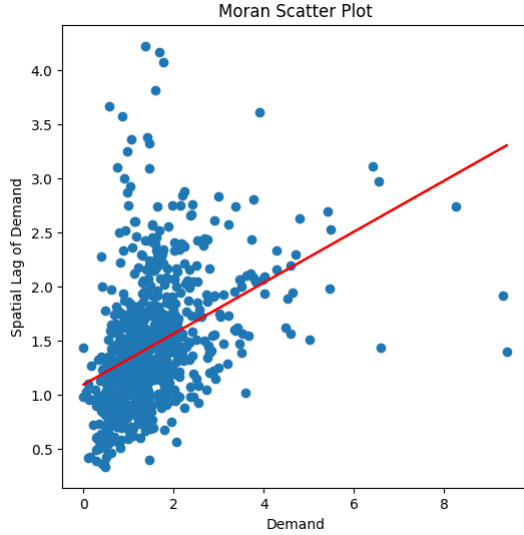


Fig. 7. Moran scatter plot showing positive spatial autocorrelation between station demand and neighboring demand.

The computed Moran's I value is positive and statistically significant, indicating that stations with high demand tend to be located near other high-demand stations, and similarly for low-demand stations.

C. Local Spatial Structure (LISA)

While global statistics indicate overall spatial dependence, Local Indicators of Spatial Association (LISA) are used to identify specific regions of clustering.

The results reveal distinct hotspots where high-demand stations are concentrated in close proximity, as well as cold spots with consistently low activity.

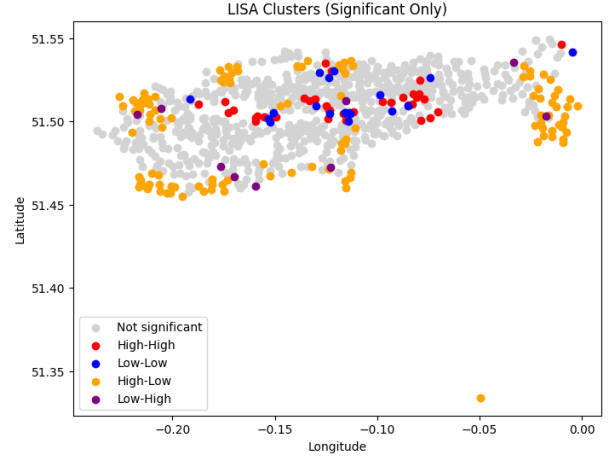


Fig. 8. LISA cluster map showing high-high (hotspots) and low-low (cold spots) regions of demand.

D. Origin-Destination Patterns

To further understand movement patterns, an origin-destination (OD) matrix is constructed by aggregating trips between pairs of stations.

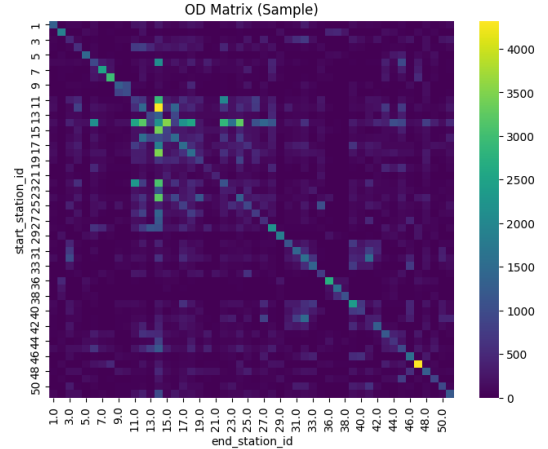


Fig. 9. Heatmap of dominant origin-destination flows between high-traffic stations.

The analysis reveals that a small number of station pairs account for a significant proportion of trips, indicating repeated travel patterns between key locations.

E. Spatial Clustering

To further explore spatial structure, K-means clustering is applied using geographic coordinates and demand values.

The combined clustering approach produces more meaningful groupings by capturing both location and activity.

F. Spatial Demand Prediction

To quantify the relationship between spatial features and demand, machine learning models are applied.

Feature importance analysis indicates that neighboring demand is the most influential predictor.

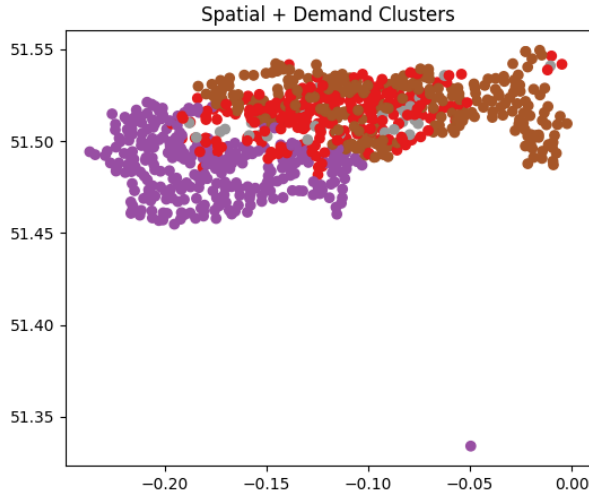


Fig. 10. Combined clustering of stations based on spatial location and demand characteristics.

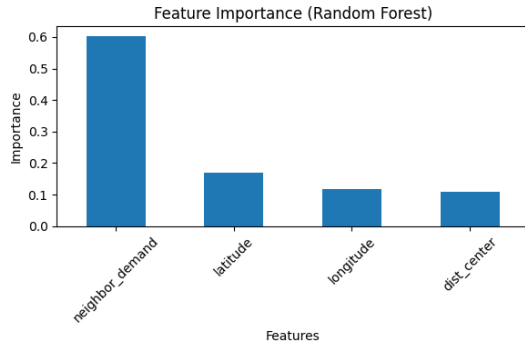


Fig. 11. Feature importance from Random Forest showing the dominance of neighborhood demand.

Overall, the spatial analysis demonstrates that bike-sharing demand exhibits significant spatial structure, characterized by clustering, local dependence, and predictable patterns.

V. DATA MINING AND MACHINE LEARNING FRAMEWORK

A. Temporal Feature Engineering and Modeling

Temporal feature engineering is performed on the aggregated hourly demand data. Key features include lag-based variables such as previous hour demand, previous day demand (24-hour lag), and previous week demand (168-hour lag), which capture temporal dependencies. Additionally, time-based features such as hour of day, day of week, and weekend indicators are incorporated to represent periodic patterns. Rolling statistics are also computed to capture short-term trends and variability.

Machine learning models, including Linear Regression, Random Forest and XGBoost are applied to predict hourly demand.

The results in Table I show that tree-based models outperform linear regression, indicating the presence of nonlinear temporal relationships. The effectiveness of these models is

attributed to strong seasonality and structured temporal behavior in the dataset.

B. Spatial Feature Engineering and Modeling

For spatial analysis, station-level demand is combined with geographic information to construct spatial features. These include longitude, latitude, distance from the city center, and neighborhood-based features such as average demand of nearby stations.

Linear Regression and Random Forest models are used to predict station-level demand.

TABLE II
RMSE COMPARISON FOR SPATIAL DEMAND PREDICTION MODELS

Model	RMSE
Linear Regression	0.7039
Random Forest	0.8123

The results indicate that linear models perform competitively and often outperform more complex models, suggesting that spatial demand relationships are relatively smooth and well-structured. The strong influence of neighboring demand highlights the importance of local spatial interactions.

C. Spatial Data Mining Tasks

Several spatial data mining techniques are applied to understand the structure of demand. Global spatial autocorrelation is measured using Moran's I and Geary's C, both of which indicate significant spatial clustering. Local patterns are identified using LISA, which reveals hotspots and cold spots of demand.

Additionally, clustering techniques are used to group stations based on geographic location, demand, and combined features. The results demonstrate that demand patterns are not purely geographic but emerge from the interaction between spatial proximity and usage behavior. Origin-destination analysis further reveals dominant travel flows between specific station pairs, indicating structured movement patterns.

D. Temporal Data Mining Tasks

Temporal data mining is performed using sequence mining techniques. Demand values are categorized into discrete levels (low, medium, high), and frequent sequential patterns are extracted.

TABLE III
MOST FREQUENT TEMPORAL DEMAND SEQUENCES (LENGTH-4)

Pattern	Frequency
LLLL	6274
MMMM	3566
HHHH	2478
HHMM	1754
HHHM	1501
MHHH	1445
HMMM	1407
LLLM	1326
MMHH	1050
MMMH	876

The results show that demand states tend to persist over consecutive time periods, indicating stability in temporal dynamics. Autocorrelation analysis further supports the presence of strong periodicity, particularly at daily intervals. These findings justify the use of lag-based features in predictive modeling.

E. Joint Spatio-Temporal Inference

Combining spatial and temporal analyses reveals that bike-sharing demand is influenced by both temporal regularity and spatial dependence. Temporal models capture recurring patterns driven by daily routines, while spatial models highlight the importance of local neighborhood interactions.

The consistency between spatial statistics (Moran's I, LISA), clustering results, and machine learning feature importance demonstrates that demand is not independent across stations or time. Instead, it emerges from the interaction of temporal behavior and localized spatial structure.

VI. TAIPEI ANALYSIS

A. Temporal Demand Patterns

To provide a comparative perspective, temporal analysis is performed on the Taipei bike-sharing dataset using the same aggregation strategy as London. The data is grouped at an hourly resolution to construct a system-level demand time series.

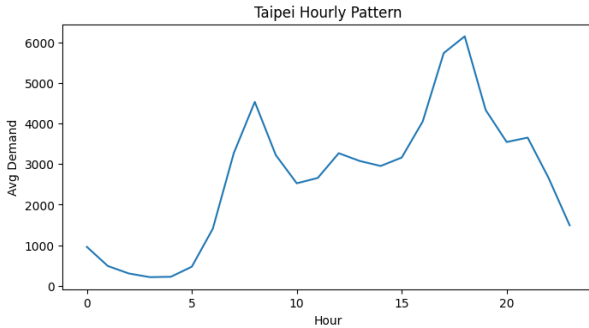


Fig. 12. Hourly bike-sharing demand in Taipei showing smoother temporal variation compared to London.

The resulting time series exhibits clear periodic behavior, with demand varying consistently across different times of the day. However, compared to London, the peaks in Taipei are less pronounced, indicating a more evenly distributed usage pattern throughout the day.

Seasonal decomposition further confirms the presence of daily patterns, though with smoother transitions and lower variability. This suggests that bike usage in Taipei is less sharply driven by commuter peaks and may reflect more flexible or continuous usage behavior.

B. Temporal Modeling

Machine learning models are applied to the Taipei time series using the same feature engineering approach as in the

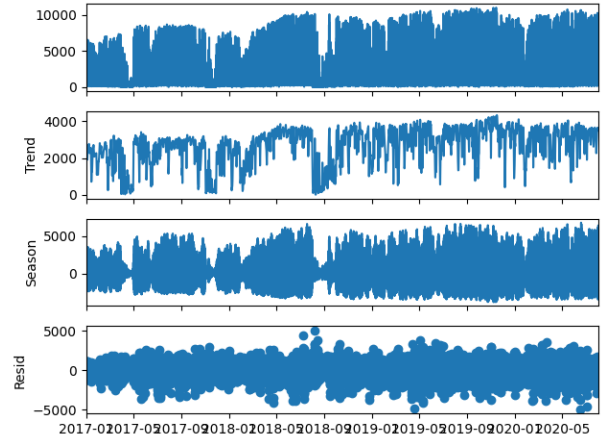


Fig. 13. STL decomposition of Taipei demand highlighting smoother seasonal patterns and lower variability.

TABLE IV
TEMPORAL MODEL PERFORMANCE FOR TAIPEI DATASET

Model	RMSE
Linear Regression	1017.897
Random Forest	495.474

London analysis. Lag-based features, along with time-of-day and day-of-week indicators, are used to predict hourly demand.

The models achieve strong predictive performance, indicating that temporal dependencies are still present and can be effectively captured. However, due to the smoother demand patterns, prediction errors are more evenly distributed and less concentrated around specific peak periods.

C. Station-Level Demand Distribution

Station-level demand is computed by averaging the number of trips per station over time. The distribution of demand across stations shows variability, with certain stations consistently exhibiting higher activity. However, compared to London, the disparity between high-demand and low-demand stations is less extreme, suggesting a more balanced usage across the network.

D. Comparison with London

A comparison between London and Taipei highlights key differences in demand behavior. London exhibits strong peak-hour patterns, driven by commuting activity, leading to sharp fluctuations in demand over time. In contrast, Taipei demonstrates smoother temporal variation and a more uniform distribution of demand across stations.

These differences indicate that bike-sharing usage is influenced by city-specific factors such as urban structure, commuting habits, and transportation infrastructure. While both cities exhibit predictable temporal patterns, the nature of demand variability differs significantly, emphasizing the importance of contextual analysis in urban mobility studies.

VII. DISCUSSION AND FUTURE WORK

While the current analysis captures key temporal and spatial patterns, several extensions could further improve understanding of bike-sharing demand.

First, incorporating external data such as weather conditions, public events, and traffic information could enhance predictive performance, as these factors are known to influence bike usage. Second, more advanced temporal models, such as recurrent neural networks, could be explored to better capture long-term dependencies and complex sequential behavior.

From a spatial perspective, graph-based models could be used to explicitly represent the network structure of stations and flows between them. Additionally, deeper analysis of origin-destination patterns and user-level trajectories could provide further insight into movement behavior.

Finally, integrating spatial and temporal modeling into a unified framework could enable more accurate forecasting and a better understanding of dynamic interactions in bike-sharing systems.

VIII. CONCLUSION

This work presented a comprehensive spatio-temporal analysis of bike-sharing systems using large-scale datasets from London and Taipei. The study demonstrated that bike demand exhibits strong temporal regularity, characterized by clear daily seasonality and peak-hour behavior driven by commuting patterns. Temporal modeling results showed that feature-based machine learning approaches can effectively capture these patterns, with nonlinear models providing improved performance due to the structured and recurring nature of demand.

From a spatial perspective, the analysis revealed significant spatial autocorrelation, indicating that station demand is not independent but influenced by neighboring stations. Statistical measures such as Moran's I and LISA confirmed the presence of clustered hotspots and cold spots, while clustering and origin-destination analysis highlighted the existence of structured spatial and movement patterns. Machine learning results further reinforced these findings, showing that local neighborhood features play a dominant role in predicting station-level demand.

The comparative analysis with Taipei highlighted differences in demand behavior across cities, with London exhibiting sharper peak-driven patterns and Taipei showing smoother and more evenly distributed usage. These differences emphasize the importance of contextual factors in urban mobility systems.

Overall, the results demonstrate that bike-sharing demand is governed by the interaction of temporal dynamics and localized spatial dependencies. Understanding this interplay is essential for accurate demand prediction and effective planning of shared mobility systems. The combined use of statistical analysis, data mining, and machine learning provides a robust framework for analyzing complex urban mobility data.

DECLARATION OF GENERATIVE AI TOOLS

Generative AI tools were used during the preparation of this assignment to assist with code debugging, clarification of spatio-temporal analysis concepts, and improvement of the structure and clarity of written explanations. These tools were also used to review sections of the report for grammatical correctness and readability.

All data preprocessing steps, implementation of the temporal and spatial analysis pipelines, feature engineering, and development of machine learning models were performed and verified by the authors. The final code, figures, experimental results, and conclusions presented in this report reflect the authors' understanding of spatio-temporal data analysis applied to bike-sharing systems in London and Taipei.

TASK OWNERSHIP AND CONTRIBUTION

The work presented in this study was carried out collaboratively, with specific components undertaken by each author as follows:

Keshav Goyal was responsible for data preprocessing, temporal analysis, feature engineering for temporal models, and implementation of machine learning models for demand prediction. This includes time series construction, STL decomposition, lag-based feature design, and evaluation of regression models.

Rishit contributed to spatial analysis, including preparation of station-level datasets, computation of spatial statistics such as Moran's I and Geary's C, and implementation of LISA for identifying local spatial patterns.

Himanshu worked on spatial data mining and visualization tasks, including clustering analysis (geographic, demand-based, and combined), origin-destination analysis, and supporting visualization of spatial patterns.

The report has been written collaboratively, with each author contributing to the sections corresponding to their respective components.

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